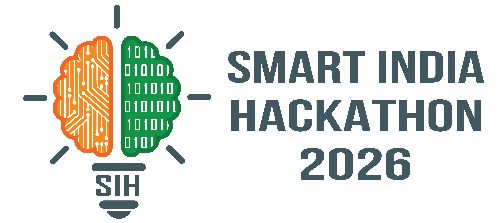


SMART INDIA HACKATHON 2026



BIS SAHAYAK

Standards guidance for consumers and MSMEs

Problem Statement ID: SIH26107

Problem Statement Title

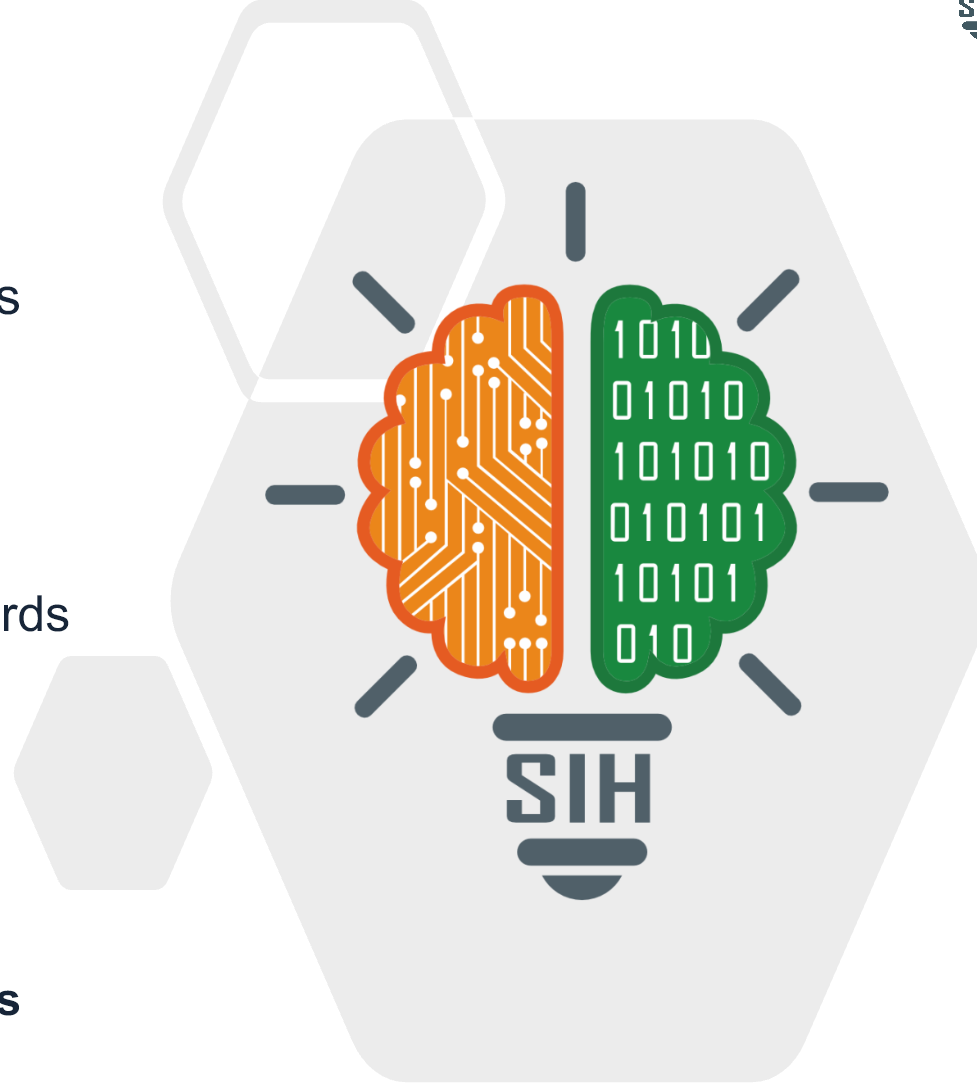
AI-powered intelligent assistant for Indian Standards and BIS services for industries and consumers

Theme: Smart Automation (confirm in SIH portal)

PS Category: Software Team Name: Logic Lords

Team ID: [Enter registered team ID]

[Live prototype: sih2026-bis-assistant.vercel.app](https://sih2026-bis-assistant.vercel.app)



Proposed solution

For a consumer

Ask: “How do I check this pressure cooker?”
Get a short answer and an official BIS Care handoff.

For an MSME

Ask: “Which tests apply to my product?”
Inspect a cited summary, then check the BIS source.

Working prototype

English or Hindi, two modes, audio, standards search and service guides.

Two users. One route to the right source.



[Try the live prototype](#)

Current scope: 21 demonstration summaries.
Official BIS portals remain the authority.

Methodology: retrieve, explain, inspect

Stage	What the prototype does	Result
1 Ask	Capture query, user mode, language and limited history	Consumer or MSME context
2 Retrieve	Rank local summaries using weighted terms and cosine similarity	Up to five relevant chunks
3 Explain	Pass selected text to Gemini 3.5 Flash-Lite; flag fallback in API	A cited answer or fallback
4 Inspect	Link citations to a demonstration standard view	Source link for official check

Implemented stack

Next.js 15 + React 19 + TypeScript + Tailwind
Local JSON retrieval, Gemini, Web Speech API, Vercel

Working prototype, with a proposed review pilot

21

curated standards

2

languages and user modes

32 / 32

automated tests passed

Risk or dependency	Control proposed for the pilot
Wrong or outdated guidance	BIS-reviewed records, version dates and evidence checks
Incomplete licence coverage	Official lookup handoff. Missing data means “unconfirmed”
API outages and rising usage cost	Fallback disclosure, quotas, caching and cost tracking

Six-week pilot plan (proposed)

Weeks 1–2: audit sources Weeks 3–4: test with users Weeks 5–6: measure and expand

Potential benefits for the people who use standards

Consumers

Understand the mark and reach official verification.
Hindi explanations and audio improve access.

MSMEs

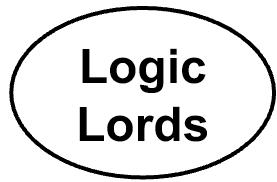
Find applicable standards and testing information.
Reduce repeated searches before certification work.

Public service teams

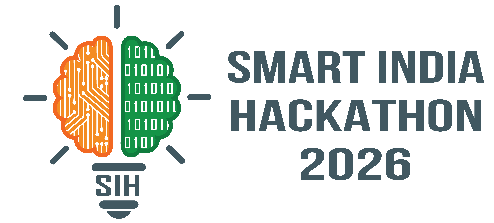
Guide routine questions to the right service.
Help users understand what to do next.

Pilot evaluation: 20 consumers + 10 MSME users

Measure	Proposed target, not yet achieved
Task completion time	50% lower median time than manual portal search
Citation precision	At least 95% relevant references on 100 reviewed queries
Safety and accessibility	No unsupported numeric limits in the reviewed set. Test both languages.



RESEARCH AND REFERENCES



Evidence base and official service links

BIS: Know Your Standard

Official standards, amendments and related services

bis.gov.in/know-your-standard

BIS Care

Official licence and hallmark verification, complaints

bis.gov.in/bis-apps

BIS laboratory search

Authoritative route for laboratory information

lims.bis.gov.in/home/search_is_number

Lewis et al., 2020

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

arxiv.org/abs/2005.11401

Project implementation

Source code, curated records and test suite

github.com/PhoenixOP231/BIS_Sahayak_SIH

Live prototype

Try a question, inspect its citation, change user mode

sih2026-bis-assistant.vercel.app